Lab 1 Report – Artificial Intelligence

1. AI Timeline

Year	Milestone	Significance
1950	Turing Test	Introduced a test of machine-like human conversation.
1956	Dartmouth Conference	Helped establish AI as a research field.
1974–1980	AI Winter	Funding and interest in AI research declined.
1997	Deep Blue	IBM computer defeated chess champion Garry Kasparov.
2012	AlexNet / GPU deep learning	Showed the impact of large datasets and GPU computing.

Sources: Turing (1950), *Computing Machinery and Intelligence*; McCarthy et al. (1955), Dartmouth proposal; Encyclopaedia Britannica, AI winter; IBM, Deep Blue history; Krizhevsky et al. (2012), *ImageNet Classification with Deep CNNs*.

2. Rule-Based Tests

Rules: score ≥ 70, documents complete, attendance ≥ 75%, and interview score ≥ 50. All thresholds are synthetic lab assumptions.

Case	Input summary	Expected
1	Score 80; complete; attendance 85%; interview 70	Pass
2	Attendance 60%; other rules pass	Fail: attendance
3	Interview 40; other rules pass	Fail: interview
4	Score 70; other rules pass	Pass: boundary

Cases were selected to show one pass, each new-condition failure, and the exact score boundary. The program's actual output should be checked against these expected results.

3. Conversational Test

The Turing Test evaluates human-like conversation; it does not prove general intelligence or factual accuracy. For example, a chatbot may confidently give incorrect scholarship eligibility advice. A reliable task test should check marks and documents against official scholarship rules.

4. Limitation and Reflection

A score of 69 fails the score rule even if other results are strong. Borderline cases should receive human review. This lab showed the need for transparent rules, testing, and human verification.